Started on	Sunday, 24 July 2022, 6:25 PM
State	Finished
Completed on	Sunday, 24 July 2022, 6:25 PM
Time taken	11 secs
Marks	0.00/95.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 5.00

History of Pharmacology (2)

For each of the following historical figures choose one most appropriate discovery

The first antidepressant agent to be discovered — Choose...

The first antipsychotic shown to have therapeutic effects — Choose...

The first depot injectable form of antipsychotic medication introduced in the market — Choose...

The first SSRI to be launched — Choose...

The first benzodiazepine to be introduced — Choose...

Choose...
Iproniazid
Chlorpromazine
Fluphenazine depots
Zimeldine
Fluoxetine
Haloperidol
Chlordiazepoxide
Diazepam
Risperidone depot
Lithium

Explanation:

The first drug used for the treatment of depression was **iproniazid** in the 1950s. The choice of this drug was based on its euphoric effects in tuberculosis patients. Subsequently, it was demonstrated that iproniazid was a monoamine-oxidase (MAO) inhibitor, particularly MAO-A.

Early in their Vietnam War (late 1940s and early 1950s), French pharmacologists employed a combination of the antihistamine phenergan and an opioid medication in the wounded, facilitating analgesia, alleviating fear, and opposing shock. While working on similar compounds, Delay and Deniker developed the first antipsychotic agent, **chlorpromazine**, at the Rhône-Poulenc Pharmaceutical Company, under the direction of Paul Charpentier. First studied by a French military surgeon for trauma, psychiatrists soon adapted it for its 'tranquilising' effects.

Fairly soon after the introduction of chlorpromazine, antipsychotics became available as long-acting injections. The first compounds were **fluphenazine** enanthate and decanoate, with development attributed to GR Daniel, a medical director at Squibb & Sons.

Zimeldine was the first of the SSRIs to reach the market. It was synthesised in 1971 and launched by the Swedish firm Astra in 1982, but subsequently withdrawn the next year because of adverse neurological effects.

In 1955, **chlordiazepoxide**, the first benzodiazepine, was synthesised by Sternbach for Roche.

Ref: Macedo D et al. Antidepressants, antimicrobials or both? Gut microbiota dysbiosis in depression and possible implications of the antimicrobial effects of antidepressant drugs for antidepressant effectiveness. Journal of affective disorders. 2017 Jan 15;208:22-32.

Westermeyer J. Cultural Psychiatry in the French-Speaking World. In Cultural Psychiatry 2013 (Vol.

33, pp. 56-63). Karger Publishers.

Crocq MA. A history of antipsychotic long-acting injections in the treatment of schizophrenia. L'encephale. 2015 Jan 15;41(1):84-92.

Mulinari S. Divergence and convergence of commercial and scientific priorities in drug development: The case of Zelmid, the first SSRI antidepressant. Social Science & Medicine. 2015 Aug 1;138:217-24.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 38.

The correct answer is: The first antidepressant agent to be discovered → Iproniazid, The first antipsychotic shown to have therapeutic effects → Chlorpromazine, The first depot injectable form of antipsychotic medication introduced in the market → Fluphenazine depots, The first SSRI to be launched → Zimeldine, The first benzodiazepine to be introduced → Chlordiazepoxide

Question **2**

Not answered

Marked out of 4.00

Adverse effects (4)

For each of the following description of adverse effects , choose the most likely offending agent.

A 40 year-old-gentleman develops prolonged painful erection on antidepressant treatment	Choose...
A 23-year-old gentleman commenced on an antipsychotic has a significantly raised prolactin	Choose...
A 20-year-old gentleman with first episode psychosis develops significant weight gain on treatment with an antipsychotic	Choose...
A 45-year-old woman upon suddenly stopping an antidepressant becomes anxious, feels electric shock sensations and becomes dizzy	Choose...

Choose...
Carbamazepine
Reboxetine
Paroxetine
Mirtazapine
Sodium valproate
Mianserin
Clozapine
Lithium
Trazodone
Amisulpride
Olanzapine

Explanation:

In rare cases, trazodone causes priapism. Trazodone should be discontinued if prolonged penile erection occurs because of the risk of permanent erectile dysfunction. Patients should seek medical attention immediately if painful erections occur lasting more than an hour. Generally, priapism reverses spontaneously, while penile blood flow and other signs are being monitored, but in urgent cases, local phenylephrine injections or even surgery may be indicated.

Amisulpride, sulpiride, risperidone, and paliperidone are particularly associated with hyperprolactinaemia.

Antipsychotics associated with the highest risk of weight gain are clozapine and olanzapine.

Discontinuation effects can be more common or more severe with paroxetine than with other SSRIs. Paroxetine's discontinuation effects may be related to the fact that it inhibits its own metabolism. Thus, when paroxetine is withdrawn, the rate of its decline can be faster as it stops inhibiting its metabolism. Readaptation of cholinergic receptors after prolonged blockade may also contribute to the discontinuation effects of paroxetine.

Ref:Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 59(794-795.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 38-39.

The correct answer is: A 40 year-old-gentleman develops prolonged painful erection on antidepressant treatment → Trazodone, A 23-year-old gentleman commenced on an antipsychotic has a significantly raised prolactin → Amisulpride, A 20-year-old gentleman with first episode psychosis develops significant weight gain on treatment with an antipsychotic → Olanzapine, A 45-year-old woman upon

suddenly stopping an antidepressant becomes anxious, feels electric shock sensations and becomes dizzy → Paroxetine

Question **3**

Not answered

Marked out of 5.00

Adverse effects (5)

Which of the drugs in the given list most likely fits the adverse effect profile described below?

An 83-year-old lady suffers constipation and dizziness, and incurs falls, on starting this antidepressant — Choose...

A 45-year-old patient with bipolar disorder on treatment develops acute abdominal pain and significantly raised serum amylase levels — Choose...

A 45-year-old gentleman with treatment-resistant schizophrenia started on a new antipsychotic develops chest pain, tachycardia, tachypnoea and palpitations — Choose...

A 55-year-old patient with chronic schizophrenia was on longstanding treatment with an antipsychotic medication that caused pigmentary retinopathy — Choose...

A 50- year-old gentleman with chronic schizophrenia is on an antipsychotic medication for a long period; he develops blue gray discolouration of skin exposed to sunlight — Choose...

Choose...
Trazodone
Venlafaxine
Amitriptyline
Fluoxetine
Olanzapine
Sodium valproate
Clozapine
Thioridazine
Carbamazepine
Chlorpromazine
Mirtazapine

Explanation:

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Its blockade of alpha adrenergic 1 receptors can cause dizziness, sedation, and hypotension. Sedative effects and weight gain can occur due to antihistaminic properties. Anticholinergic activity can cause sedative effects, dry mouth, constipation, and blurred vision.

Rarely, valproate can cause pancreatitis, which is sometimes fatal.

Clozapine is associated with cardiomyopathy. It may occur at any time. Cardiomyopathy should be suspected in any patient showing signs of heart failure, which should provoke immediate cessation of clozapine. Presentation of cardiomyopathy varies somewhat and is often asymptomatic in the early stages, so any reported symptoms of palpitations, chest pain, syncope, sweating, decreased exercise capacity or breathing difficulties should be closely investigated.

High-dose chlorpromazine can result in abnormal pigmentation of the eyelids, cornea, and lens. However, this is not associated with visual disturbance and resolves on cessation of treatment. Thioridazine (now withdrawn) was also associated with pigmentary retinopathy. Corneal oedema can also (rarely) occur with chlorpromazine, leading to loss of vision if treatment is not withdrawn.

The long-term use of phenothiazines or tricyclic antidepressants may induce hyperpigmentation, usually in areas exposed to the sun. Chlorpromazine is undoubtedly the drug most frequently implicated, causing violet or metallic grey purple pigmentation in areas exposed to the sun, mainly the face and limbs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 25 26, 830.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 222-223.

Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 519.

García RM, Molina SC. Drug-induced hyperpigmentation: review and case series. The Journal of the

American Board of Family Medicine. 2019 Jul 1;32(4):628-38.

The correct answer is: An 83-year-old lady suffers constipation and dizziness, and incurs falls, on starting this antidepressant → Amitriptyline, A 45-year-old patient with bipolar disorder on treatment develops acute abdominal pain and significantly raised serum amylase levels → Sodium valproate, A 45-year-old gentleman with treatment-resistant schizophrenia started on a new antipsychotic develops chest pain, tachycardia, tachypnoea and palpitations → Clozapine, A 55-year-old patient with chronic schizophrenia was on longstanding treatment with an antipsychotic medication that caused pigmentary retinopathy → Thioridazine, A 50- year-old gentleman with chronic schizophrenia is on an antipsychotic medication for a long period; he develops blue gray discolouration of skin exposed to sunlight → Chlorpromazine

Question **4**

Not answered

Marked out of 4.00

Choose...
Fluoxetine
Zopiclone
Oxazepam
Nitrazepam
Paroxetine
Esketamine
Lamotrigine
Imipramine
Diazepam
Lithium
Sodium valproate

Pharmacokinetics (2)

Match each characteristic below with the most likely drug from the list above;

This drug undergoes N-dealkylation during phase 1 metabolism, which results in desmethyl derivative that has a longer half-life than the parent compound	Choose...
This drug undergoes direct phase 2 metabolism without undergoing phase 1 metabolism	Choose...
The active metabolite of this drug has an elimination half-life of 1-2 weeks	Choose...
This drug has narrow therapeutic index	Choose...

Explanation:

Most benzodiazepines are oxidised first by CYP3A4 and CYP2C19, often to active metabolites. These metabolites may then be hydroxylated to another active metabolite. For example, diazepam is oxidised to desmethyldiazepam, which, in turn, is hydroxylated to produce oxazepam. These products undergo glucuronidation to inactive metabolites. Several benzodiazepines (e.g., diazepam, chlordiazepoxide) have the same active metabolite (desmethyldiazepam), which has an elimination half-life of more than 120 hours.

It is not essential that a drug must undergo phase 1 metabolism via CYP system in order to undergo phase 2 metabolism. For example, lorazepam, temazepam and oxazepam undergo direct phase 2 reactions. They are conjugated directly by glucuronidation and are excreted. As a result, in patients with alcoholic liver disease, oxazepam is a favoured drug for alcohol detoxification instead of chlordiazepoxide.

Fluoxetine has a long half-life (2-3 days), and its active metabolite an even longer half-life (2 weeks). In total, fluoxetine may take up to 4 -5 weeks to be cleared from the body substantially.

Some drugs have a narrow therapeutic index (a small increase in dose can cause toxicity and a small reduction in dose can cause a loss of therapeutic actions). The most commonly prescribed ones are digoxin, warfarin, theophylline, phenytoin, and lithium.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 670.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 602, 780.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 293.

The correct answer is: This drug undergoes N-dealkylation during phase 1 metabolism, which results

in desmethyl derivative that has a longer half-life than the parent compound → Diazepam, This drug undergoes direct phase 2 metabolism without undergoing phase 1 metabolism → Oxazepam, The active metabolite of this drug has an elimination half-life of 1-2 weeks → Fluoxetine, This drug has narrow therapeutic index → Lithium

Question **5**

Not answered

Marked out of 4.00

Effect of receptor action (3)

Match the receptor below with its most significant effect from the list above

Consequences of stimulating 5HT1A receptor — Choose...

Consequences of stimulating 5HT2 receptors — Choose...

Consequences of stimulating 5HT3 receptors — Choose...

Decreased brain GABA function — Choose...

Choose...
Orthostatic hypotension
Therapeutic effects of antidepressants
Production of negative symptoms
Withdrawal symptoms caused by MAOIs
Withdrawal symptoms from benzodiazepines
Excessive weight gain
Gastrointestinal side-effects of SSRIs
Withdrawal symptoms caused by SSRIs
Leads to features of hyperprolactinaemia
Leads to extra-pyramidal side effects
Insomnia and sexual side effects of SSRIs

Explanation:

Serotonin action at 5HT1A receptors facilitates downstream norepinephrine, dopamine, and acetylcholine release. Many psychotropic drugs that treat psychosis, mood, and anxiety are 5HT1A agonists or partial agonists. The stimulation of the 5HT1A receptor leads to its therapeutic effects.

It is thought that stimulation of 5HT2A receptors has negative effects on sexual function. Antidepressants drugs that increase 5HT signalling in the absence of 5HT2A antagonism (e.g. SSRIs) are commonly associated with sexual side effects while agents that possess 5HT2A antagonist activity (e.g mirtazapine) have a relatively low likelihood of inducing sexual side effects.

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs). Blocking these 5HT3 receptors can therefore protect against chemotherapy-induced nausea and vomiting as well as against serotonin-induced gastrointestinal side effects that can accompany agents that increase serotonin.

Benzodiazepines belong to a class of drugs that modulate the neurotransmitter GABA, especially GABA-A, to open the chloride channel. Benzodiazepines have a powerfully calming effect on the brain. They can modulate GABA-A receptors as an agonist. GABA has a fantastic property of being an inhibitory neurotransmitter with the ability to reduce the excitability of neurons. Benzodiazepine use can cause physical dependence and tolerance. After a period of 1-6 months of use, the abrupt cessation of benzodiazepines can cause life-threatening seizures, delirium, and death related to a reduction in GABA functions.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021.

Osis L, Bishop JR. Pharmacogenetics of SSRIs and sexual dysfunction. Pharmaceuticals. 2010 Dec;3(12):3614-28.

Fluyau D et al. Challenges of the pharmacological management of benzodiazepine withdrawal, dependence, and discontinuation. Therapeutic advances in psychopharmacology. 2018 May;8(5):147-68.

The correct answer is: Consequences of stimulating 5HT1A receptor → Therapeutic effects of antidepressants, Consequences of stimulating 5HT2 receptors → Insomnia and sexual side effects of SSRIs, Consequences of stimulating 5HT3 receptors → Gastrointestinal side-effects of SSRIs,

Decreased brain GABA function → Withdrawal symptoms from benzodiazepines

Question **6**

Not answered

Marked out of 4.00

Side effects and receptors (4)

Choose one explanatory option for each of the following conditions:

Late onset Alzheimer disease — Choose...

Clozapine induced constipation — Choose...

SSRI-induced nausea — Choose...

Post-injection delirium with olanzapine — Choose...

Choose...
Dopamine D4 activation
Muscarinic antagonism
Serotonin 5-HT2 receptor blockade
Serotonin 5HT3 blockade
Apolipoprotein E receptors
High 5HT2 serotonin receptor sensitivity
Serotonin 5HT3 activation
Low D3 receptor occupancy in striatum
Decreased nicotinic receptor density
Low dopamine receptor sensitivity

Explanation:

Apolipoprotein E (ApoE) is a lipid carrier in both periphery and the central nervous system. Lipid-loaded ApoE lipoprotein particles bind to several cell surfaces to support membrane homeostasis and injury repair in the brain. Considering prevalence and relative risk magnitude, the E4 allele of the APOE gene is the strongest genetic risk factor for late-onset Alzheimer disease.

The pharmacological basis of clozapine-related constipation is multifactorial. The drug's anticholinergic and antihistaminergic properties, along with an antagonism at 5-HT3 receptors contributes directly to reduced gastrointestinal motility. Mean transit times are four times longer than normal and 80% of clozapine patients show reduced transit time. Also, the sedative effects of clozapine can also indirectly contribute to this by reducing physical activity.

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs). Blocking these 5HT3 receptors (e.g., using ondansetron) can therefore protect against chemotherapy-induced nausea and vomiting as well as against serotonin-induced gastrointestinal side effects that can accompany agents that increase serotonin.

Olanzapine produces muscarinic M1-5 receptor antagonism, explaining its anticholinergic-like effects. Accidental excess entry of injected olanzapine to bloodstream (intravasation) causes a surge in its effects leading to post-injection syndrome. It is characterised by a range of features starting from mild sedation to outright confusion, delirium and coma. By blocking histamine H1 receptors it produces sedation, while orthostatic hypotension results from the blockade of adrenergic α1 receptors.

Ref: Zhao N et al Apolipoprotein E, receptors, and modulation of Alzheimer's disease. Biological psychiatry. 2018 Feb 15;83(4):347-57.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 232.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 309-311.

The correct answer is: Late onset Alzheimer disease → Apolipoprotein E receptors, Clozapine induced constipation
→ Serotonin 5HT3 blockade, SSRI-induced nausea → Serotonin 5HT3 activation, Post-injection delirium with olanzapine → Muscarinic antagonism

Question **7**

Not answered

Marked out of 4.00

Pharmacodynamics (3)

Choose the drug that best fits the description below

This drug weakly blocks the synaptic reuptake of dopamine	Choose...
This drug blocks 5-HT2C receptors	Choose...
This SSRI drug has an anticholinergic action	Choose...
This drug is a reversible inhibitor of monoamine oxidase A	Choose...

Choose...
Sertraline
Fluvoxamine
Dothiepin
Venlafaxine
Fluoxetine
Paroxetine
Escitalopram
Esketamine
Citalopram
Moclobemide

Explanation:

Sertraline has two possible mechanisms that distinguish it from other SSRIs: dopamine transporter (DAT) inhibition and sigma-1 receptor binding. The DAT inhibitory actions are controversial since they are weaker than the serotonin transporter (SERT) inhibitory actions. However, it is not clear that high degrees of DAT occupancy are necessary or even desirable in order to contribute to antidepressant actions. That is, perhaps only a small amount of DAT inhibition is sufficient to cause improvement in energy, motivation, and concentration, especially when added to another action such as SERT inhibition.

In addition to serotonin reuptake inhibition, fluoxetine also has 5HT2C antagonistic actions that may explain many of its unique clinical properties. 5HT2C antagonism may contribute to its antidepressant actions and also to its efficacy in other disorders, especially in eating disorders. Other drugs for unipolar depression with 5HT2C antagonist properties include trazodone, mirtazapine, agomelatine, and some tricyclic antidepressants.

In addition to serotonin reuptake inhibition, paroxetine has mild anticholinergic actions (M1), which can be calming or possibly sedating.

Moclobemide is a reversible inhibitor of monoamine oxidase-A (RIMA).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 293-294.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 282-283.

The correct answer is: This drug weakly blocks the synaptic reuptake of dopamine → Sertraline, This drug blocks 5-HT2C receptors
→ Fluoxetine, This SSRI drug has an anticholinergic action → Paroxetine, This drug is a reversible inhibitor of monoamine oxidase A → Moclobemide

Question **8**

Pharmacodynamics (4)

Choose the drug that best fits the description below.

This drug carries the least risk of pharmacodynamic drug interactions among SSRIs	Choose...
A preferential blockade of dopamine transporter is seen with this drug	Choose...
At high doses of administration of this drug, there is a blockade of the reuptake of serotonin, noradrenaline and dopamine	Choose...
This drug preferentially blocks noradrenaline reuptake with a negligible serotonergic effect and a safe cardiovascular profile	Choose...

Choose...
Escitalopram
Reboxetine
Paroxetine
Mirtazapine
Citalopram
Fluoxetine
Fluvoxamine
Moclobemide
Bupropion
Venlafaxine

Explanation:

Escitalopram is considered perhaps the best-tolerated SSRI, with the fewest CYP450-mediated drug interactions.

The prototypical norepinephrine-dopamine reuptake inhibitor is bupropion. Bupropion has weak blocking properties for the dopamine transporter (DAT) and for the norepinephrine transporter (NET). No other specific or potent pharmacological actions have been consistently identified for this agent, including on serotonin. Bupropion's antidepressant actions may be explained in part by the more potent inhibitory properties of its metabolites. Bupropion is metabolised to a number of active metabolites, some of which are not only more potent NET inhibitors than bupropion itself and equally potent DAT inhibitors, but are also concentrated in the brain.

Venlafaxine boosts neurotransmitters serotonin, norepinephrine/noradrenaline, and dopamine. It blocks the serotonin reuptake pump (serotonin transporter - SERT), presumably increasing serotonergic neurotransmission. It blocks the norepinephrine reuptake pump (norepinephrine transporter – NET), presumably increasing noradrenergic transmission. Since dopamine is inactivated by norepinephrine reuptake in the frontal cortex, which largely lacks dopamine transporters, venlafaxine can increase dopamine neurotransmission in this part of the brain. Venlafaxine also weakly blocks the dopamine reuptake pump (dopamine transporter – DAT), and may increase dopamine neurotransmission.

Reboxetine is a selective norepinephrine reuptake inhibitor. It has relatively low cardiotoxicity. It has no effect on QTc, but rhythm abnormalities may occur.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 296, 303-304.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 841.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 384, 390.

Antidepressant	Effect on QTc
Agomelatine	Single case of QTc prolongation
Bupropion	QTc shortening, but prolongation has been reported in cases of overdose
Citalopram/escitalopram	Dose-related increase in QTc
Duloxetine	Isolated reports of prolongation
Fluoxetine	No effect on QTc interval
Fluvoxamine	No significant effect on QTc
Lofepramine	Can possibly prolong QTc interval at higher doses
MAOIs	Unclear but may shorten QTc interval
Mirtazapine	No effect on QTc
Moclobemide	No effect on QTc interval in normal doses; prolongation in overdose
Paroxetine	No effect on QTc interval
Reboxetine	No effect on QTc
Sertraline	No effect on QTc interval
Trazodone	Can prolong QTc interval
Tricyclics	Prolongation of QTc interval and QRS interval
Venlafaxine/desvenlafaxine	Possible prolongation in overdose, but very rare
Vilazodone	No effect, even in overdose
Vortioxetine	No effect
	Adapted from Maudsley Prescribing Guidelines, 2018

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The correct answer is: This drug carries the least risk of pharmacodynamic drug interactions among SSRIs → Escitalopram, A preferential blockade of dopamine transporter is seen with this drug → Bupropion, At high doses of administration of this drug, there is a blockade of the reuptake of serotonin, noradrenaline and dopamine → Venlafaxine, This drug preferentially blocks noradrenaline reuptake with a negligible serotonergic effect and a safe cardiovascular profile → Reboxetine

Question **9**

Not answered

Choose...
Duloxetine
Aripiprazole
Reboxetine
Molindone
Fluoxetine
Bupropion
Galantamine
Agomelatine
Paliperidone
Buspirone

Mechanism of action (2)
Find the drug that matches each of the following mechanisms of action

Selective norepinephrine reuptake inhibitor	Choose...
Melatonin agonist	Choose...
D2 partial agonist	Choose...
5-HT1A partial agonist	Choose...

Explanation:

Reboxetine is a selective norepinephrine reuptake inhibitor.

Agomelatine has agonist actions at melatonin 1 (MT1) and melatonin 2 (MT2) receptors and antagonist actions at 5HT-2C and 5HT-2B receptors.

Aripiprazole is a D2 receptor partial agonist. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors.

Buspirone is classified as an azapirone (or azaspirone) and is chemically distinct from other psychotropic agents. It is a 5-HT1A partial agonist.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 681

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 306.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 672.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 176.

The correct answer is: Selective norepinephrine reuptake inhibitor → Reboxetine, Melatonin agonist → Agomelatine, D2 partial agonist → Aripiprazole, 5-HT1A partial agonist → Buspirone

Question **10**

Not answered

Marked out of 8.00

Choose...
5-HT3 agonism
5-HT3 antagonism
H2- antagonism
D2 stimulation
5-HT1A stimulation
H3- antagonism
D2 blockade
5-HT2A stimulation
H1 antagonism
5-HT2C stimulation

Receptor action and effects (5)
For the following side effects exhibited by the drugs, identify the causative receptor action.

A 45-year-old gentleman develops akathisia and restlessness on treatment with risperidone	Choose...
A 45-year-old gentleman develops insomnia on taking paroxetine	Choose...
A 25-year-old gentleman complains of delayed ejaculation on treatment with fluoxetine	Choose...
A 56-year-old woman complains of gaining weight on treatment with amitriptyline	Choose...
A 49-year-old lady has shown improvement on her depressive symptoms by taking sertraline for more than four weeks	Choose...

Explanation:

The side effects of antipsychotic medications like akathisia, EPSEs and hyperprolactinaemia are caused primarily by D2 receptor blockade. Akathisia and agitation are also induced by 5HT2A stimulation (hence some data on mirtazapine- 5HT2A antagonist reducing akathisia).

Patients on SSRIs may develop insomnia due to 5-HT2A stimulation and can develop delayed ejaculation due to 5HT2A/2C stimulation.

The weight gain induced by psychotropics is due to antihistaminic effects and 5HT2C antagonism.

5HT1A stimulation by SSRIs such as sertraline indirectly produces antidepressant action

The correct answer is: A 45-year-old gentleman develops akathisia and restlessness on treatment with risperidone → D2 blockade, A 45-year-old gentleman develops insomnia on taking paroxetine → 5-HT2A stimulation, A 25-year-old gentleman complains of delayed ejaculation on treatment with fluoxetine → 5-HT2C stimulation, A 56-year-old woman complains of gaining weight on treatment with amitriptyline → H1 antagonism, A 49-year-old lady has shown improvement on her depressive symptoms by taking sertraline for more than four weeks → 5-HT1A stimulation

Question **11**

Not answered

Marked out of 4.00

Addiction Pharmacology

Choose the drug that best fits descriptions below

Aldehyde dehydrogenase inhibitor	Choose...
Non-selective long-acting opiate antagonist	Choose...
Mu opioid receptor partial agonist	Choose...
Shorter acting mu opioid antagonist	Choose...

Choose...
Disulfiram
Oxycodone
Naltrexone
Naloxone
Chlordiazepoxide
Acamprosate
Oxazepam
Buprenorphine

Explanation:

Disulfiram is an irreversible inhibitor of the liver enzyme aldehyde dehydrogenase that normally metabolises alcohol.

Naltrexone is a non-selective opioid receptor antagonist.

Buprenorphine is a partial agonist at mu-opioid receptors.

Naloxone is an opioid receptor antagonist that can reverse opioid overdose. Naloxone has a much shorter half-life than naltrexone.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 556.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 463, 474, 478, 496.

The correct answer is: Aldehyde dehydrogenase inhibitor → Disulfiram, Non-selective long-acting opiate antagonist → Naltrexone, Mu opioid receptor partial agonist → Buprenorphine, Shorter acting mu opioid antagonist → Naloxone

Question **12**

Not answered

Adverse effects (3)

Identify the side effect most specifically attributed to each drug.

Marked out of 5.00

Flag

Choose...
Hair loss
Stevens-Johnson Syndrome
Tics
Agranulocytosis
Worsening of psoriasis
Amenorrhoea
Hepatitis
Weight loss
Pruritus
Hepatomegaly
Anticholinergic effects

Paroxetine	Choose...
Lithium	Choose...
Amitriptyline	Choose...
Lamotrigine	Choose...
Sodium valproate	Choose...

Explanation:

In addition to serotonin reuptake inhibition, paroxetine has mild anticholinergic actions (M1), which can be calming or possibly sedating.

Some skin conditions such as psoriasis and acne can be aggravated by lithium therapy.

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Anticholinergic activity can cause sedative effects, dry mouth, constipation, blurred vision, and confusion.

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g. Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with a number of systemic manifestations).

Hair loss (with curly regrowth) can occur with valproate.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 294.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 250, 259.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 25-26.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 358.

The correct answer is: Paroxetine → Anticholinergic effects, Lithium → Worsening of psoriasis, Amitriptyline → Anticholinergic effects, Lamotrigine → Stevens-Johnson Syndrome, Sodium valproate → Hair loss

Question 13

Not answered

Marked out of

Choose...
Lorazepam
Methylphenidate
Acamprosate
Mirtazapine
Lamotrigine
Carbamazepine
Phenelzine
Citalopram
Lithium
Sildenafil
Clozapine

Adverse effects (4)

For each of the adverse effects given below, identify the best drug associated with it

Insomnia	Choose...
Pedal oedema	Choose...
Blue tinged vision	Choose...
Delayed orgasm	Choose...
Rash	Choose...
Fine tremor	Choose...

Explanation:

Insomnia is common with methylphenidate.

Peripheral oedema is a notable side effect of phenelzine.

Sildenafil is considered to be generally safe. Acute secondary effects of sildenafil include headache, heartburn, skin flush, and vision changes such as blurred vision, photophobia, and cyanopsia (i.e., blue-tinted vision).

Citalopram can cause dose-dependent sexual dysfunction. In men, this includes delayed ejaculation and erectile dysfunction. In both men and women, this can include decreased sexual desire and anorgasmia.

10-14% of patients receiving lamotrigine will develop rash. Most are benign. A minority may be serious/life-threatening skin reactions requiring hospitalisation (e.g. Steven-Johnson syndrome, toxic epidermal necrolysis, angio-oedema, and a rash associated with a number of systemic manifestations).

Common side effects of lithium include mild gastrointestinal upset, fine tremor, polyuria, and polydipsia.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 250, 569.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 160, 61

Karaarslan C. Ocular Side Effects of Sildenafil That Persist Beyond 24 h—A Case Series. Frontiers in neurology. 2020 Feb 7;11:67.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 358.

The correct answer is: Insomnia → Methylphenidate, Pedal oedema → Phenelzine, Blue tinged vision → Sildenafil, Delayed orgasm → Citalopram, Rash → Lamotrigine, Fine tremor → Lithium

Question **14**

Not answered

Marked out of 7.00

Agonistic agents (1)

Match the following receptors with the drugs which have an agonistic effect on the receptors

Dopamine D2 receptors — Choose...

Mu receptors — Choose...

Sigma receptors — Choose...

Alpha-1 isoform benzodiazepine receptors — Choose...

Choose...
Fluvoxamine
Sildenafil
Carbamazepine
Isocarboxazid
Sodium valproate
Clozapine
Diazepam
Oxycodone
Ropinirole
Zolpidem

Explanation:

Ropinirole is a dopamine agonist (D2).

Most strong opioids, including morphine and oxycodone, are mu-receptor agonists.

Like sertraline, fluvoxamine binds at sigma-1 sites, but this action is more potent for fluvoxamine than for sertraline. The physiological function of sigma-1 sites is still a mystery, and thus sometimes called the 'sigma enigma', but has been linked both to anxiety and psychosis. Preclinical studies suggest that fluvoxamine may be an agonist at sigma-1 receptors, and that this property may contribute an additional pharmacological action to help explain fluvoxamine's well-known anxiolytic properties.

Zolpidem is an alpha-1 isoform selective agonist of GABA-A/benzodiazepine receptors. It selectively binds to a subtype of the benzodiazepine receptor, the alpha-1 isoform.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 476.

Nakamura A et al. Differential activation of the μ-opioid receptor by oxycodone and morphine in pain-related brain regions in a bone cancer pain model. British journal of pharmacology. 2013 Jan;168(2):375-88.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 295.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 871.

The correct answer is: Dopamine D2 receptors → Ropinirole, Mu receptors → Oxycodone, Sigma receptors → Fluvoxamine, Alpha-1 isoform benzodiazepine receptors → Zolpidem

Question **15**

Not answered

Marked out of 5.00

Flag question

Agonistic agents (2)

Match the following agonists with their neurochemical systems in the list.

Oxycodone	Choose...
Zolpidem	Choose...
Bromocriptine	Choose...
Pramipexole	Choose...
Aripiprazole	Choose...

Choose...
Mu receptor
NMDA
Serotonin 2A and Dopamine 1 receptor
Sigma
Dopamine 1 receptor
Dopamine 2 receptor
Serotonin 3
Serotonin 1A and Dopamine 2 receptor
GABA-A
Histamine-1

Explanation:

Most strong opioids, including morphine and oxycodone, are mu-receptor agonists.

Zolpidem is an alpha-1 isoform selective agonist of GABA-A/benzodiazepine receptors. It selectively binds to a subtype of the benzodiazepine receptor, the alpha-1 isoform.

Bromocriptine is an agonist to the D2 receptor and an antagonist to the D1 receptor.

Pramipexole is a dopamine D2/D3 receptor-preferring agonist commonly used to treat Parkinson disease.

Aripiprazole is a D2 receptor partial agonist. It is also a partial agonist at 5-HT1A receptors, a high affinity antagonist at 5-HT2A receptors, and a low-/moderate affinity antagonist at H1 and alpha-1 receptors.

Ref: Nakamura A et al. Differential activation of the μ-opioid receptor by oxycodone and morphine in pain-related brain regions in a bone cancer pain model. British journal of pharmacology. 2013 Jan;168(2):375-88.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 871.

Lieberknecht V et al. Pramipexole, a dopamine D2/D3 receptor-preferring agonist, prevents experimental autoimmune encephalomyelitis development in mice. Molecular neurobiology. 2017 Mar;54(2):1033-45.

Wilson SM et al. Classics in Chemical Neuroscience: Pramipexole. ACS Chemical Neuroscience. 2020 Jul 29;11(17):2506-12.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 212.

The correct answer is: Oxycodone → Mu receptor, Zolpidem → GABA-A, Bromocriptine →

Dopamine 2 receptor, Pramipexole → Dopamine 2 receptor, Aripiprazole → Serotonin 1A and Dopamine 2 receptor

Question **16**

Not answered

Marked out of 5.00

Flag question

Post-marketing surveillance

For each of the following drugs, identify the adverse effects originally reported during post-marketing surveillance of the agents.

Sertindole	Choose...
Mianserin	Choose...
MAOIs	Choose...
Thalidomide	Choose...
Zimeldine	Choose...

Choose...
Sudden cardiac death
Guillain-Barre syndrome
Phocomelia
Hepatotoxicity
Aplastic anemias
Agranulocytosis
Cheese reactions
Blood dyscrasias
PR prolongation on ECG

QTc prolongation risk is associated with several antipsychotic drugs, belonging both to typical and atypical classes (e.g. sertindole, thioridazine, droperidol, and haloperidol). Serious ventricular arrhythmias such as torsades de pointes and sudden cardiac death can result. In 1998, withdrawal of sertindole from the UK market followed after several sudden cardiac deaths. In 2001, sertindole was reintroduced in Europe under the restriction that treated patients would be enrolled in a study focusing on mortality. The results showed that sertindole had a significantly increased risk of cardiac mortality in comparison with risperidone.

Mianserin is an antidepressant with a mode of action similar to TCAs, but with fewer anticholinergic side effects. It has fewer cardiovascular problems, but blood dyscrasias are more common, especially in the elderly.

The most worrisome side effect of MAOIs is the tyramine-induced hypertensive crisis (‘cheese reaction’). The amino acid tyramine is normally transformed via GI metabolism. However, MAOIs inactivate GI metabolism of dietary tyramine, thus allowing intact tyramine to enter the circulation. A hypertensive crisis may subsequently occur because of a powerful pressor effect of the amino acid.

Thalidomide was released in the 1950s as a nonaddictive, nonbarbiturate sedative by the German pharmaceutical company, Chemie-Grunenthal. It was soon discovered to also be an effective anti-emetic and used to treat morning sickness in pregnant women. Soon, reports surfaced of patients developing peripheral neuropathy and severe birth defects in children born to mothers who tool thalidomide. It was not until 1961 that thalidomide was confirmed to be the cause of huge numbers (over 10 000) of severe birth defects (including phocomelia – a severe shortening of the limb/s due to long bones being reduced or missing and leaving distal elements in place). It was subsequently withdrawn from the UK in Nov 1961 and by 1962 from most of the world.

Zimeldine was the first of the SSRIs to reach the market. It was synthesised in 1971 and launched by the Swedish firm Astra in 1982, but subsequently withdrawn the next year because of adverse neurological effects such as Guillain-Barre syndrome (a potentially fatal neurological disorder). It was later estimated that at least 1 in 10 000 patients treated with zimeldine would develop Guillain-Barre, compared with a ‘spontaneous’ occurrence rate of 1 in 50 000. To this day how the drug causes such effects remains unclear, although an immunological process unrelated to the drug’s serotonin reuptake blocking effect seems likely.

Ref: Oteri A et al. Prescribing pattern of antipsychotic drugs during the years 1996–2010: a population-based database study in Europe with a focus on torsadogenic drugs. British journal of clinical pharmacology. 2016 Aug;82(2):487-97.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 289.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 645.

Vargesson N. Thalidomide-induced teratogenesis: History and mechanisms. Birth Defects Research Part C: Embryo Today: Reviews. 2015 Jun;105(2):140-56.

Mulinari S. Divergence and convergence of commercial and scientific priorities in drug development: The case of Zelmid, the first SSRI antidepressant. Social Science & Medicine. 2015 Aug 1;138:217-24.

The correct answer is: Sertindole → Sudden cardiac death, Mianserin → Blood dyscrasias, MAOIs → Cheese reactions, Thalidomide → Phocomelia, Zimeldine → Guillain-Barre syndrome

Question **17**

Not answered

Marked out of 5.00

Receptor mechanisms (2)

From the descriptions below identify the pharmacological terminology referred to in the given list

These are agents that produce a maximal response — Choose...

These are agents with a ceiling effect — Choose...

These agents bind to the same receptor but produce the opposite pharmacological effect — Choose...

These alter the receptor site in some way so increasing the dose of the agonist drug can reverse the effects only partially. — Choose...

These agents not only shift the curve to the right but also reduce the maximum effect — Choose...

Choose...
Full agonist
Semi agonist
Irreversible Antagonist
Potency
Affinity
Partial agonist
Efficacy
Non-competitive antagonist
Inverse agonist
Competitive antagonist

Explanation:Full agonists produce a maximal response. Examples are benzodiazepines at GABA-A complex, Bromocriptine for dopamine. Partial agonists cannot elicit a maximal response and are less effective than full agonists. Examples are Aripiprazole, buspirone and buprenorphine. Hence, these partial agonists have a ceiling effect. An inverse agonist is an agent that binds to the same receptor but produces the opposite pharmacological effect. Antagonists are drugs that interact with receptors to interfere with their activation by neurotransmitter or other agonistic molecules. If the dose of the agonist drug is increased, the effect of competitive antagonism can be overcome almost completely. Thus competitive antagonists can only reduce the potency but not the efficacy of agonists. Examples of competitive antagonism include atropine at muscarinic receptors and propranolol at beta-adrenergic receptors. The effects of a noncompetitive antagonist can only be partially reversed by increasing the dose of the agonist drug as these agents alter the configuration of the receptor site. Non-competitive antagonism reduces both the potency and the efficacy of agonists. Therefore, non-competitive antagonists not only shift the curve to the right but also reduce the maximum effect. For example, ketamine and phencyclidine are noncompetitive NMDA antagonists. Irreversible antagonists bind irreversibly to the target site e.g. most traditional MAOIs. Body cells must freshly synthesize the blocked enzyme before a releasing effect from the drug can be produced.

The correct answer is: These are agents that produce a maximal response → Full agonist, These are agents with a ceiling effect → Partial agonist, These agents bind to the same receptor but produce the opposite pharmacological effect → Inverse agonist, These alter the receptor site in some way so increasing the dose of the agonist drug can reverse the effects only partially. → Non-competitive antagonist, These agents not only shift the curve to the right but also reduce the maximum effect → Non-competitive antagonist

Question **18**

Not answered

Marked out of 4.00

Receptor mechanisms (4)

For the following side effects exhibited by the drugs, identify the receptor action responsible for it

A 40-year-old woman on fluoxetine developed nausea and vomiting	Choose...
A 65-year-old woman complained of excessive sedation on taking trazodone	Choose...
A 33-year-old gentleman suffered from dry mouth on taking amitriptyline	Choose...
A 23-year-old woman has developed weight gain and is overly sedated on taking mirtazapine	Choose...

Choose...
M2 receptor blockade
H1- antagonism
H2- antagonism
H3- antagonism
5-HT3 antagonism
5-HT3 agonism
M1 receptor blockade
5-HT2 agonism
5-HT1 agonism
5-HT1 antagonism
M3 receptor blockade

Explanation:

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs).

Trazodone blocks the serotonin transporter at antidepressant doses. It also has serotonin 2A and 2C, H1 histamine, and alpha-1-adrenergic antagonist properties, which can make it very sedating. At low doses, trazodone does not adequately block serotonin reuptake but retains its other properties.

Amitriptyline is a tricyclic antidepressant that acts as a serotonin and norepinephrine reuptake inhibitor. Anticholinergic activity due to M3 receptor blockade can cause sedative effects, dry mouth, constipation, and blurred vision.

Mirtazapine's H1 antagonistic properties are associated with sedation and weight gain.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 309-311, 535.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 25-26.

The correct answer is: A 40-year-old woman on fluoxetine developed nausea and vomiting → 5-HT3 agonism, A 65-year-old woman complained of excessive sedation on taking trazodone → H1- antagonism, A 33-year-old gentleman suffered from dry mouth on taking amitriptyline → M3 receptor blockade, A 23-year-old woman has developed weight gain and is overly sedated on taking mirtazapine → H1- antagonism

Question **19**

Not answered

Marked out of 3.00

Flag question

Side effects and receptors

Choose one explanatory option for each of the following adverse effects .

Antipsychotic-induced Parkinsonism	Choose...
Tardive dyskinesia	Choose...
SSRI-induced nausea	Choose...

Choose...
High Dopamine receptor sensitivity
Serotonin 5HT3 blockade
Decreased nicotinic receptor density
Dopamine D4 activation
Serotonin 5HT3 activation
High 5HT2 Serotonin receptor sensitivity
High D2 receptor occupancy in striatum
Low Dopamine receptor sensitivity

Explanation:

Motor side effects are caused by D2 antagonists/partial agonists blocking D2 receptors in the

nigrostriatal motor pathway. When D2 receptors are blocked acutely in the nigrostriatal pathway, this can cause drug-induced parkinsonism (DIP).

One theory explaining tardive dyskinesia (TD) is that nigrostriatal D2 receptors most sensitive to blockade trigger a form of undesirable neuroplasticity called super-sensitivity in reaction to D2 receptor blockade.

5HT3 receptors are localised in the chemoreceptor trigger zone of the brainstem, where they mediate nausea and vomiting (especially in response to cancer chemotherapy), and in the gastrointestinal tract itself where they mediate nausea, vomiting, and diarrhea/bowel motility when stimulated by serotonin (including when stimulated by serotonin that is a side effect of peripherally increased serotonin by SSRIs/SNRIs).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 165, 170, 309-311.

The correct answer is: Antipsychotic-induced Parkinsonism → High D2 receptor occupancy in striatum, Tardive dyskinesia → High Dopamine receptor sensitivity, SSRI-induced nausea → Serotonin 5HT3 activation

Question **20**

Not answered

Drugs & mechanism of action

Select one mechanism of action for each of the following drugs

Choose...
Presynaptic alpha-2 agonist
GABA-B agonist
MAO-B Inhibitor
GABA-A agonist
D-2 receptor partial agonist
Selective norepinephrine reuptake inhibitor
Inhibitor of norepinephrine and serotonin re-uptake
5HT1A partial agonist
NMDA receptor antagonist
MAO-A Inhibitor
Acetylcholinesterase inhibitor

Rivastigmine	Choose...
Selegiline	Choose...
Diazepam	Choose...
Reboxetine	Choose...
Duloxetine	Choose...

Explanation:

Rivastigmine is a cholinesterase inhibitor. It is both an acetylcholinesterase inhibitor and a butyrylcholinesterase inhibitor.

Oral selegiline is a selective MAO-B inhibitor. In the transdermal form, selegiline is a tissue-selective MAO inhibitor (MAO-A and MAO-B inhibitor in the brain and relatively selective MAO-B inhibitor in the gut).

Benzodiazepines, such as diazepam, belong to a class of drugs that modulate the neurotransmitter GABA, especially GABA-A, to open the chloride channel. Benzodiazepines have a powerfully calming effect on the brain. They can modulate GABA-A receptors as an agonist. GABA has a fantastic property of being an inhibitory neurotransmitter with the ability to reduce the excitability of neurons.

Reboxetine is a selective norepinephrine reuptake inhibitor.

Serotonin-norepinephrine reuptake inhibitors (SNRIs) combine robust serotonin transporter (SERT) inhibition with varying degrees of inhibition of the norepinephrine transporter (NET). Duloxetine is an SNRI that is characterised pharmacologically by slightly more potent SERT than NET inhibition.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 68 699, 705.

Fluyau D et al. Challenges of the pharmacological management of benzodiazepine withdrawal,

dependence, and discontinuation. Therapeutic advances in psychopharmacology. 2018 May;8(5):147-68.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 298-302.

The correct answer is: Rivastigmine → Acetylcholinesterase inhibitor, Selegiline → MAO-B Inhibitor, Diazepam → GABA-A agonist, Reboxetine → Selective norepinephrine reuptake inhibitor, Duloxetine → Inhibitor of norepinephrine and serotonin re-uptake

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